

JMGV_VD100

Active IR Stereo Camera Product Specification



1. Product Overview

1.1. Product Description

JMGV_VD100 active structured light binocular camera is a stereo vision camera deeply customized and developed according to the needs of robots and industrial stereo vision application scenarios. The industry-leading 940nm infrared structured light speckle projector is used to supplement light and texture, and OV9282 global shutter sensor has the highest tolerance in the industry for motion speed and environment. It can be widely used in industrial sorting, AGV and service robot obstacle avoidance, high precision visual positioning and other scenarios.

1.2. Product Features

- 640x400@30fps, high resolution and frame rate, widely used in various scenarios.
- Industry leading wide FOV : 92.5°x80.5°x55.2° (DxHxV) for wider sensing range.
- 940nm infrared structured light projector has extremely high environmental tolerance and can be used indoor and outdoor, dark and light environments. Compared with structured light and TOF solutions, the material of detected object has a wider applicable range.
- Adopts a global shutter sensor, which can be adapted to high speed and dark light environments.
- Dedicated ASIC binocular depth calculation chip, which consumes low power and does not occupy the computing power of the host computer;
- Type C USB 2.0 interface with locking function, high reliability and good applicability.
- Ultra-low noise, very suitable for obstacle avoidance applications
- Mature Linux and ROS 2 development SDK;

1.3. Usage Precautions

- This product is a precision device, and it is necessary to avoid collisions, drops, and vibrations to prevent the motherboard components from falling off or internal damage to the lens assembly, causing functional problems in the product or affecting its performance. It is recommended that shock absorption and collision protection be considered in the structure when applying
- Pay attention to keeping the lens clean to avoid affecting the normal operation of the camera.
- The product will generate heat after being used for a period of time, which is normal. It is recommended that the structure consider enhancing heat dissipation.
- Please do not modify or disassemble this product in any way to avoid damage to the product and loss of accuracy.
- The laser meets Class 1 standards and can be used with confidence.

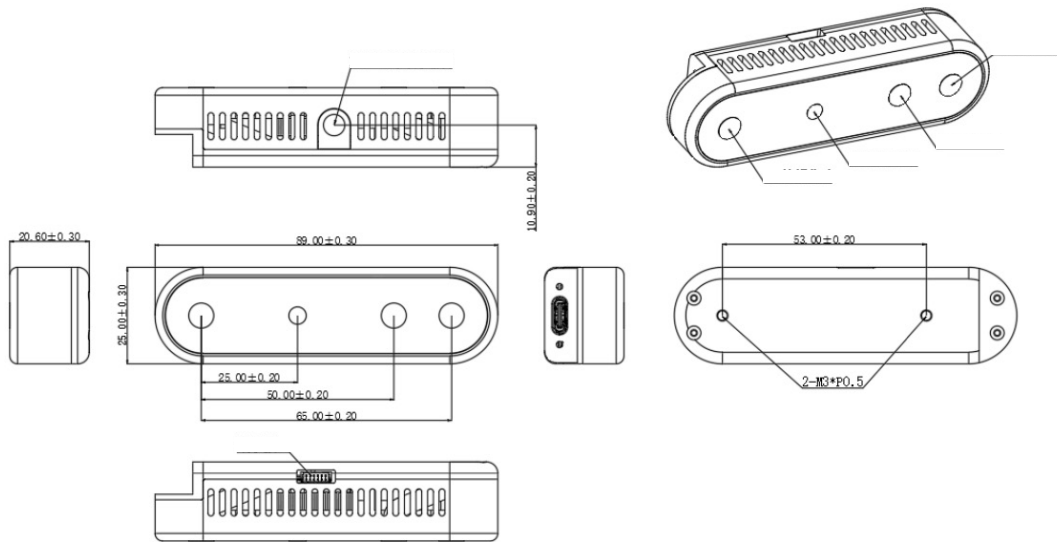
2. Product Specifications

Item	Specifications
Laser Transmitter	VCSEL 940nm , FOV: 102°x78° (HxV) No. of pixels: 30K
Infrared Camera Module	IR=940nm single-channel, FOV: 92.5°x80.5°x55.2°(DxHxV)
Transmission interface	USB2.0 interface & 6Pin terminal interface
Working Range	20cm-10m (vary depending on the environment)
Depth Map Resolution	640x400@30fps
Depth Accuracy	1m≤1% , 2m≤2%
Depth Map Fill rate	≥98%
Spatial Noise	≤2%
Working Current	Peak Value: 1A (TX&RX Operating Simultaneously), Average 260mA
Input Voltage	DC 5.0V
Power Consumption	Average power consumption: 1W, peak power consumption: 5.45W
Supported Operating Systems	Ubuntu 20.04&22.04
Operating Temperature	Operating temperature: -25°C to 70°C; recommended operating temperature: -10°C to 60°C
Working Distance	Indoor: 10m, Outdoor: 5m
Laser Safety	Class 1
Main Product Dimensions	89.0x25.0x20.6 mm

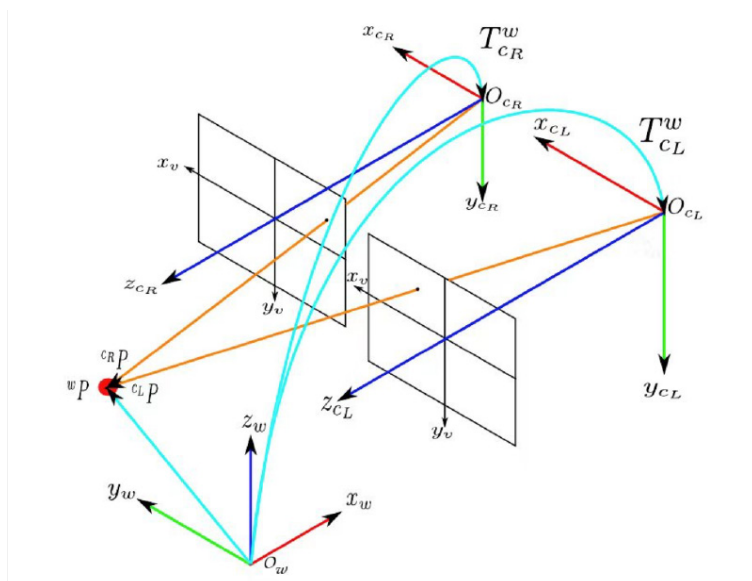
3. Interface Definition

Pin No.	1	2	3	4	5	6
SIGNAL	VCC_5V	RX	TX	GND	NC	NC

4. Product Structure

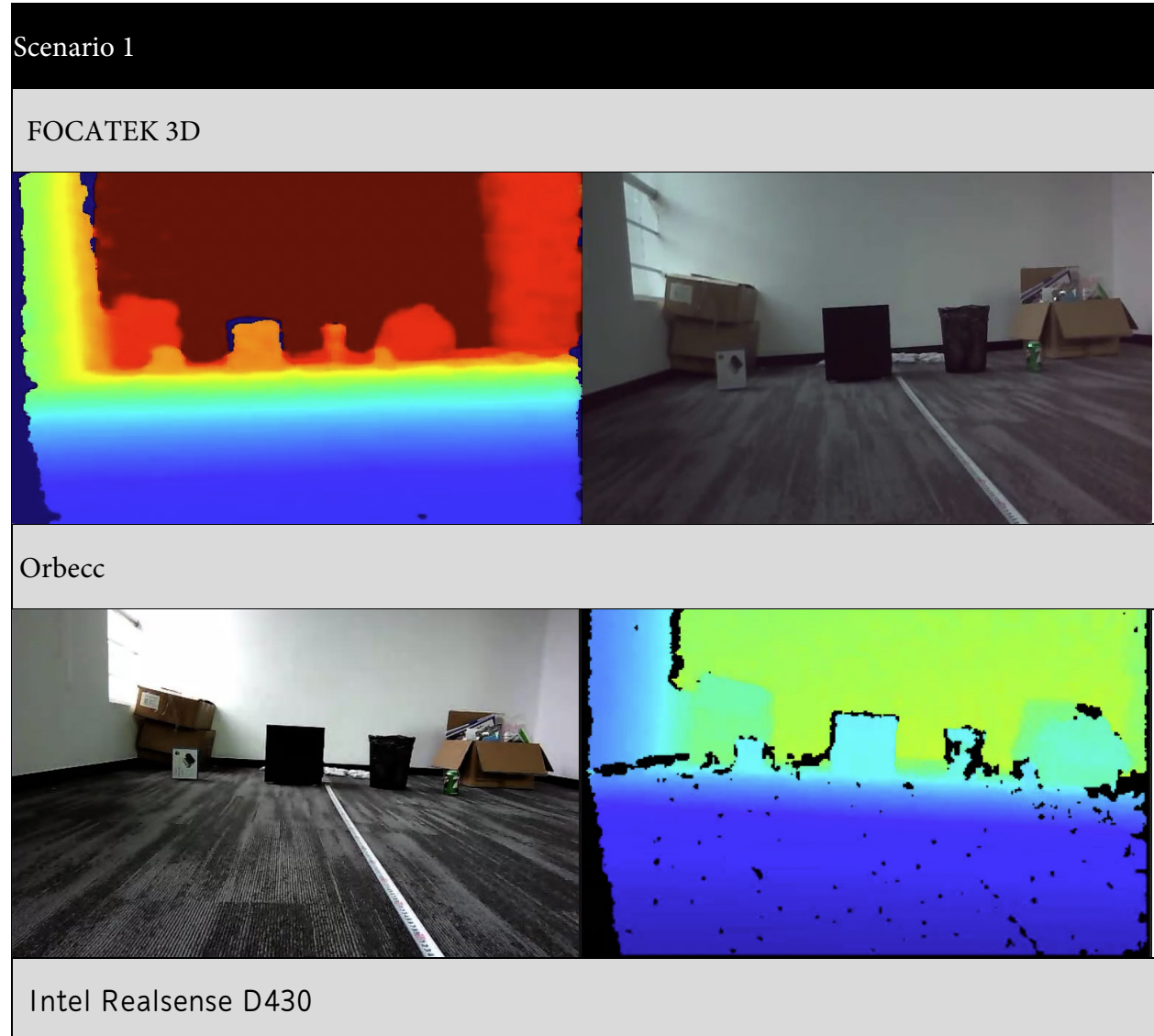


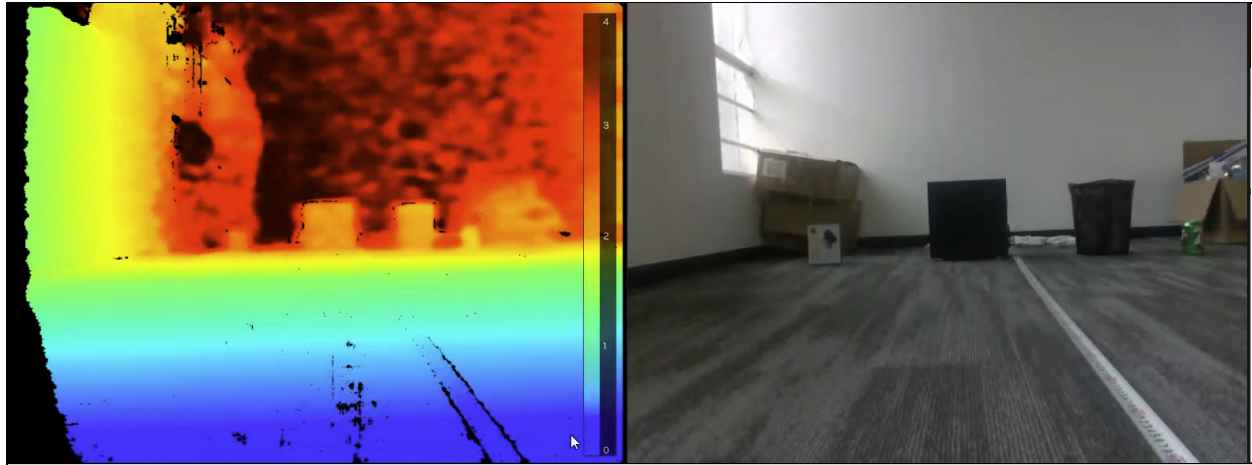
5. Spatial Coordinate System



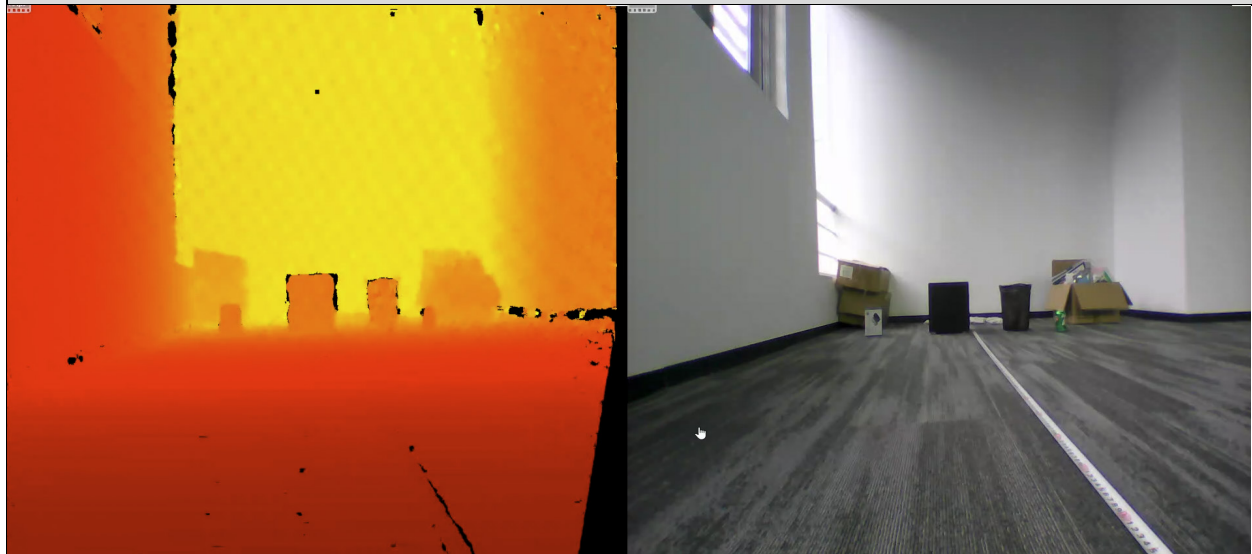
6. Comparison of Depth Effects - Competitors

Using the camera's built-in ASIC chip, and comparing depth effect images with three competing products: Realsense, Orbbec (Dabai), and Yinniu, in the same environment, as





Yinniu (China Brand)



7. Depth Map Specification

Our product's design is optimized for depth field of view, with a specified area (80% of the FOV) considered the region of interest, and emphasizes aligning this ROI with the expected usage and optical parameters to maintain consistency.